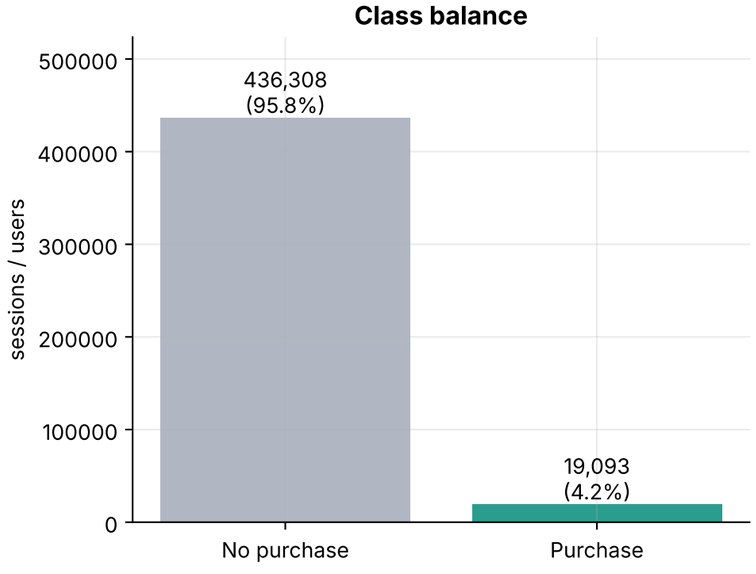
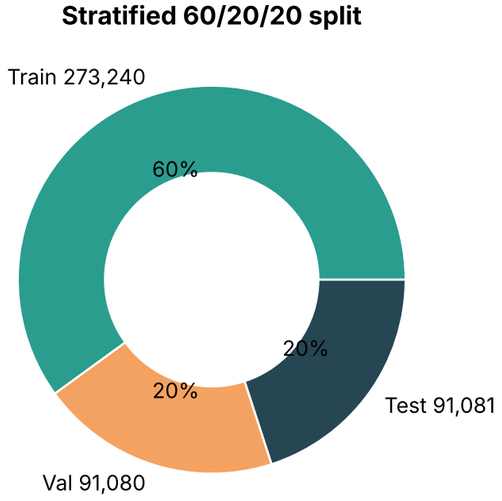
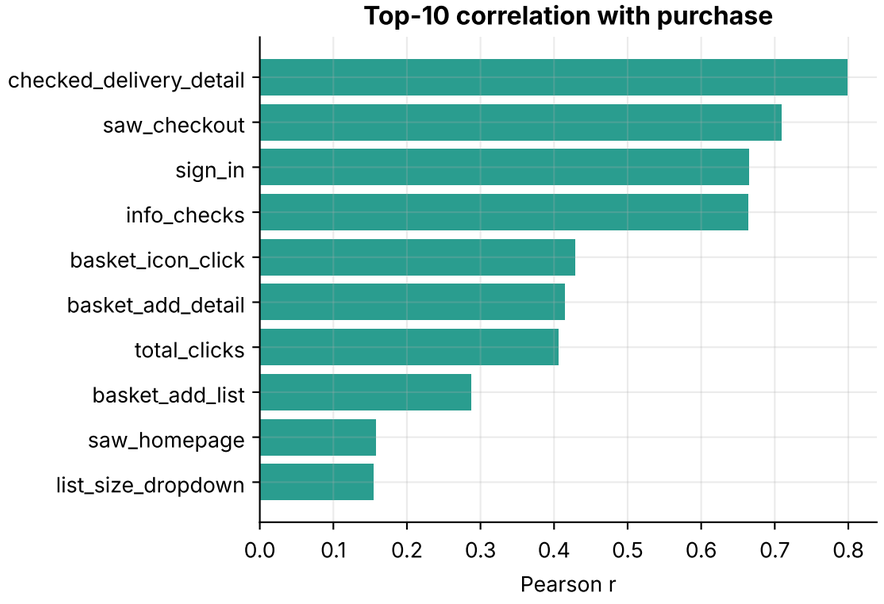
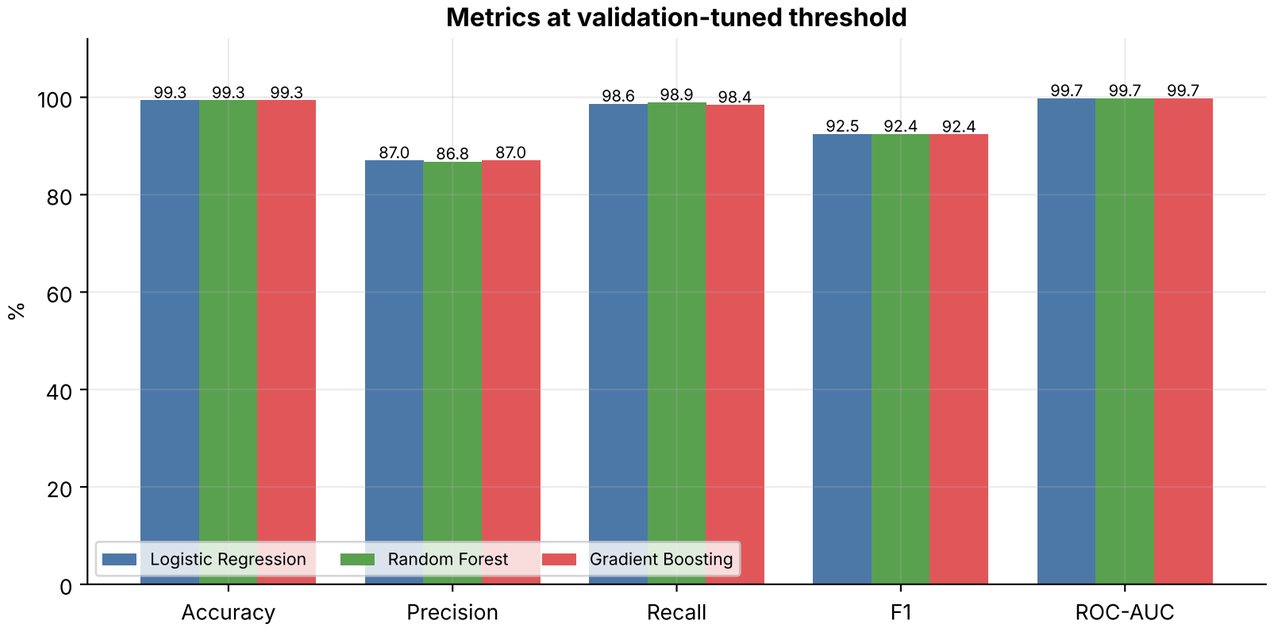
Dataset 1 (Clickstream, 'ordered') - data overview

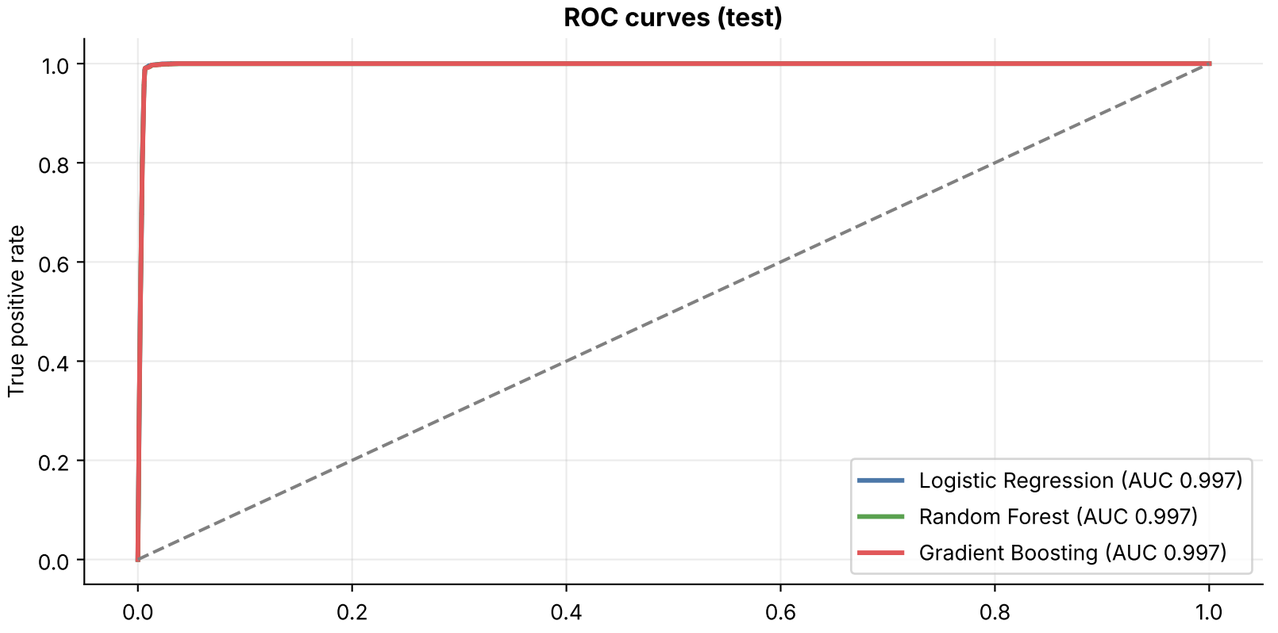


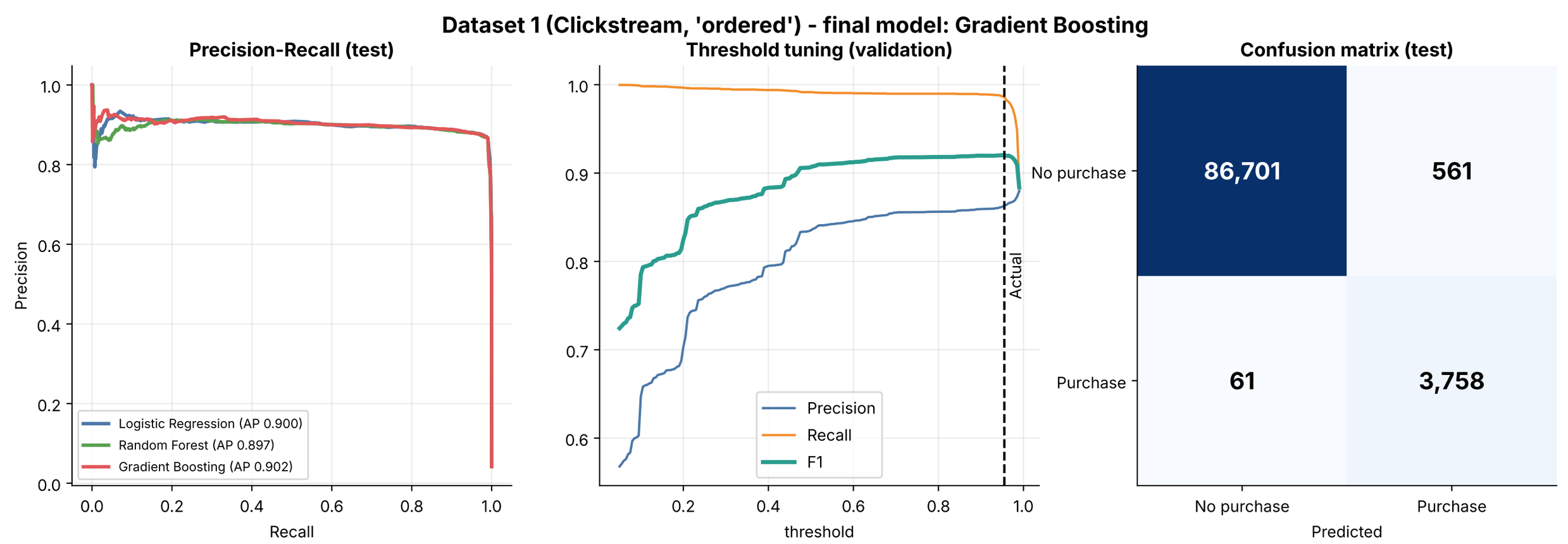


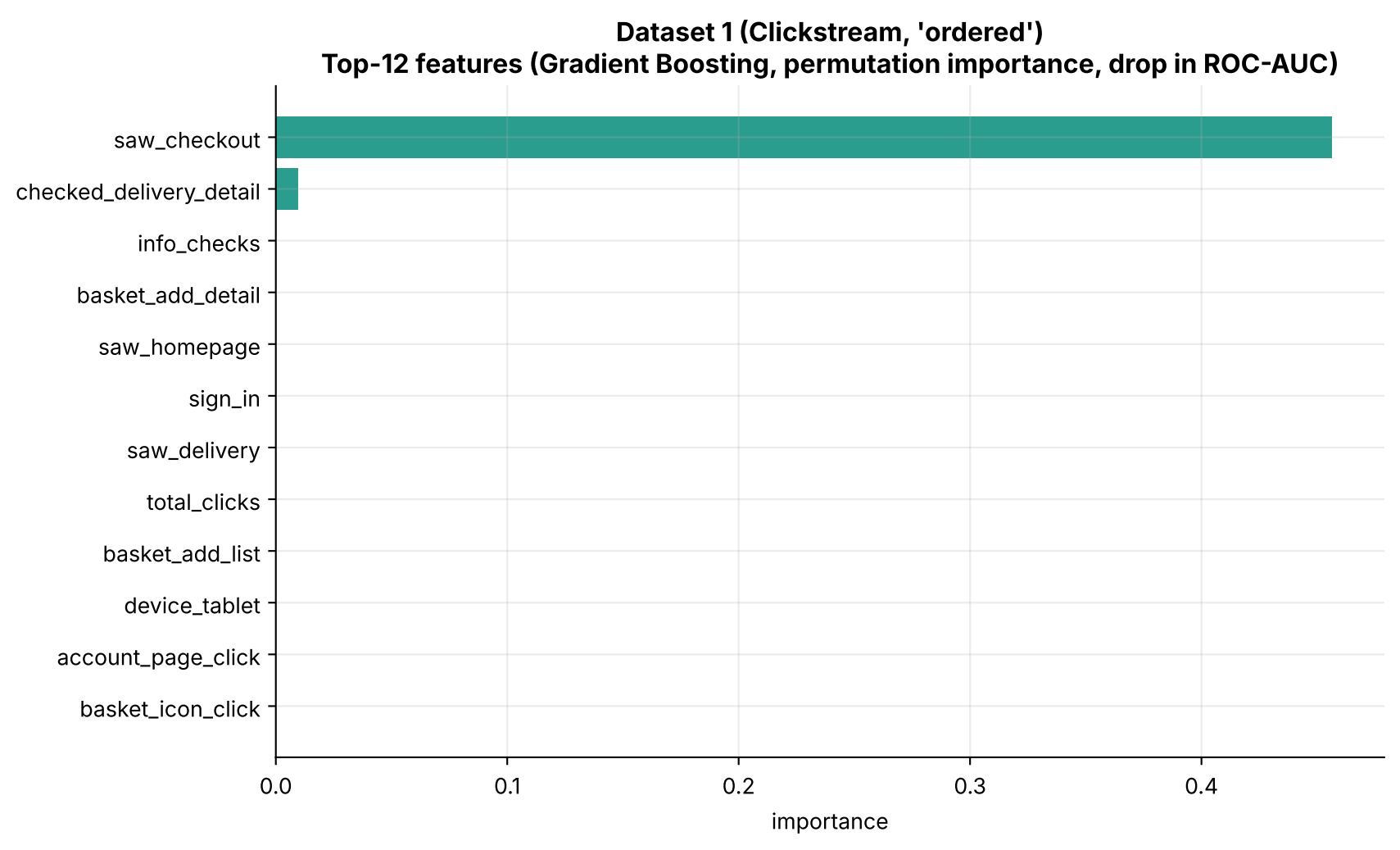


Dataset 1 (Clickstream, 'ordered') - evaluation of all models (held-out test)



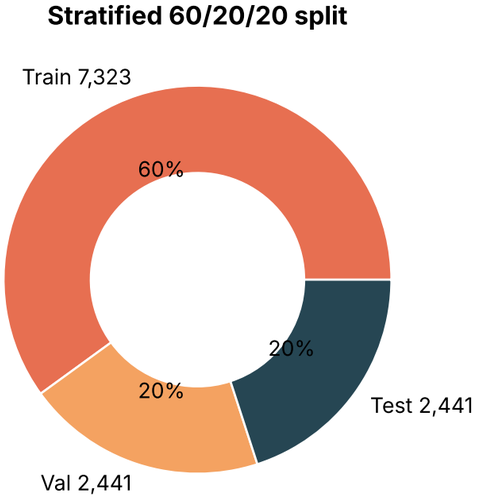


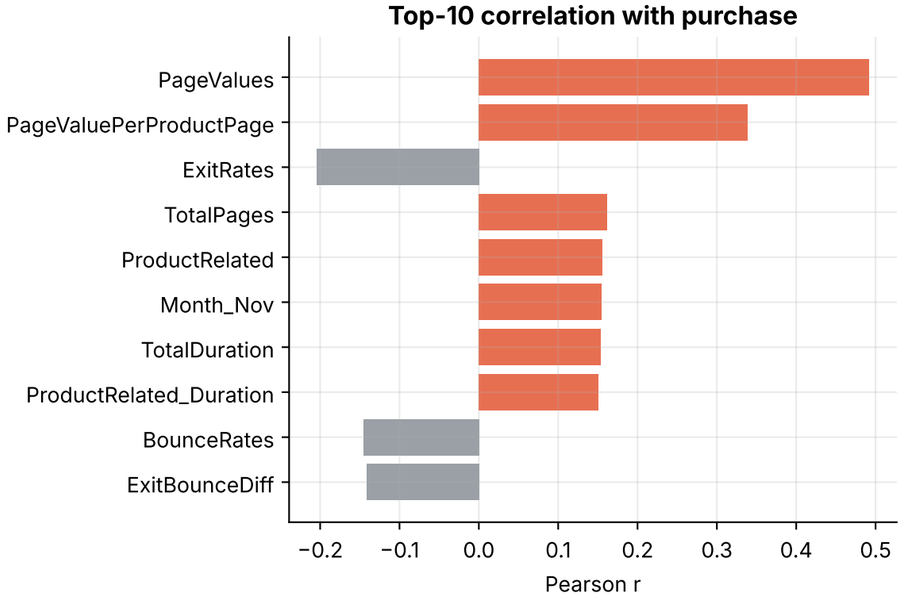

Dataset 1 (Clickstream, 'ordered') - final model: Gradient Boosting
Precision-Recall (test)
Precision
Recall
Logistic Regression (AP 0.900)
Random Forest (AP 0.897)
Gradient Boosting (AP 0.902)
Threshold tuning (validation)
threshold
Precision
Recall
F1
Actual
chosen = 0.95
Confusion matrix (test)
No purchase
Purchase
86,701
561
61
3,758
No purchase
Purchase
Predicted


Dataset 1 (Clickstream, 'ordered')
Top-12 features (Gradient Boosting, permutation importance, drop in ROC-AUC)
saw_checkout
checked_delivery_detail
info_checks
basket_add_detail
saw_homepage
sign_in
saw_delivery
total_clicks
basket_add_list
device_tablet
account_page_click
basket_icon_click
0.0
0.1
0.2
0.3
0.4
importance

Dataset 2 (UCI Online Shoppers, 'Revenue') - data overview

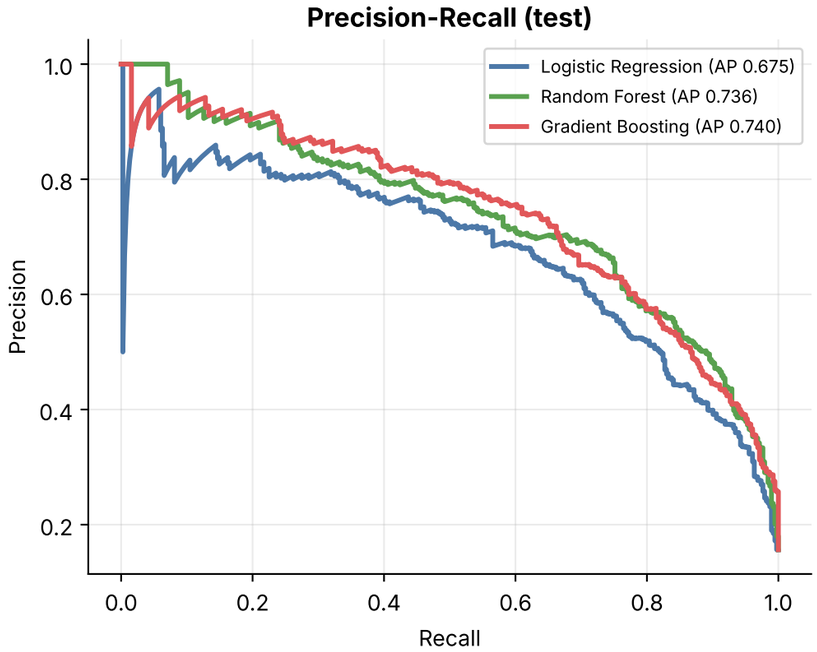


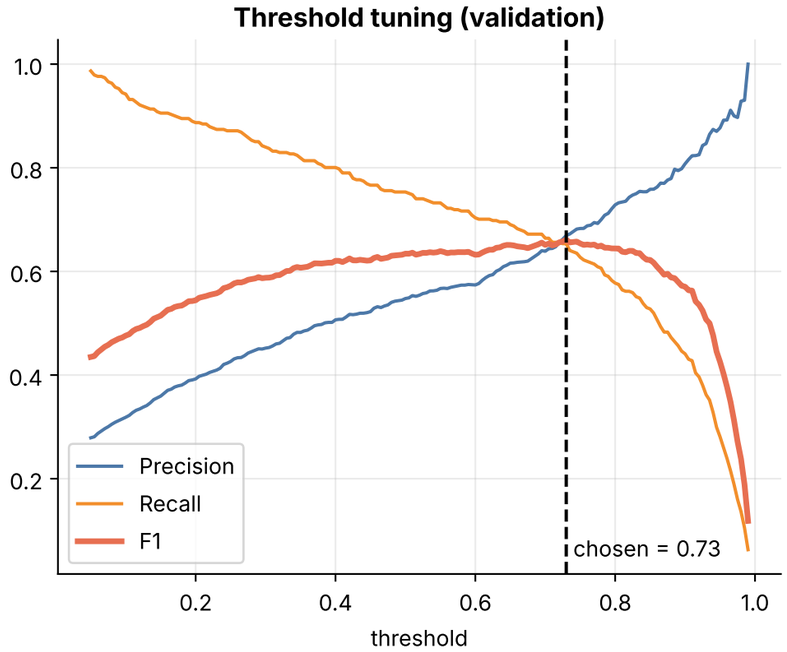


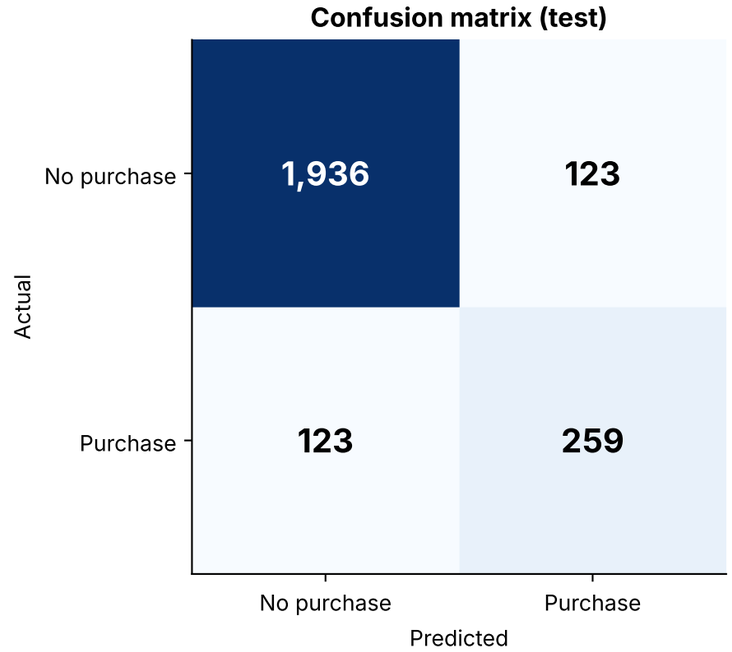


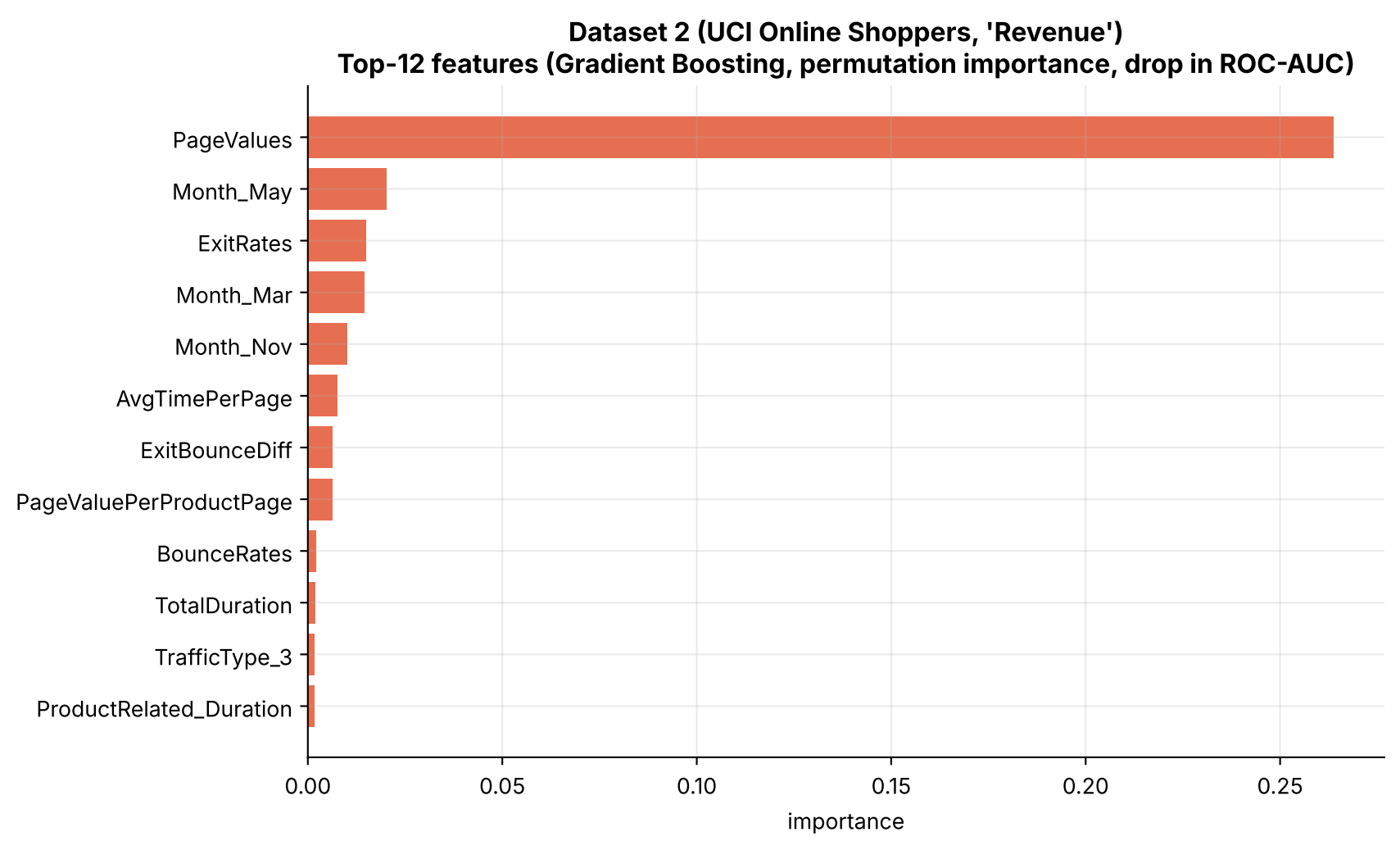

Dataset 2 (UCI Online Shoppers, 'Revenue') - evaluation of all models (held-out test)
Metrics at validation-tuned threshold
%
100
80
60
40
20
0
89.0
89.9
89.9
64.7
70.9
67.8
65.7
59.9
67.8
65.2
65.0
67.8
91.0
93.1
93.3
Logistic Regression
Random Forest
Gradient Boosting
Accuracy
Precision
Recall
F1
ROC-AUC
ROC curves (test)
True positive rate
False positive rate
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC 0.910)
Random Forest (AUC 0.931)
Gradient Boosting (AUC 0.933)

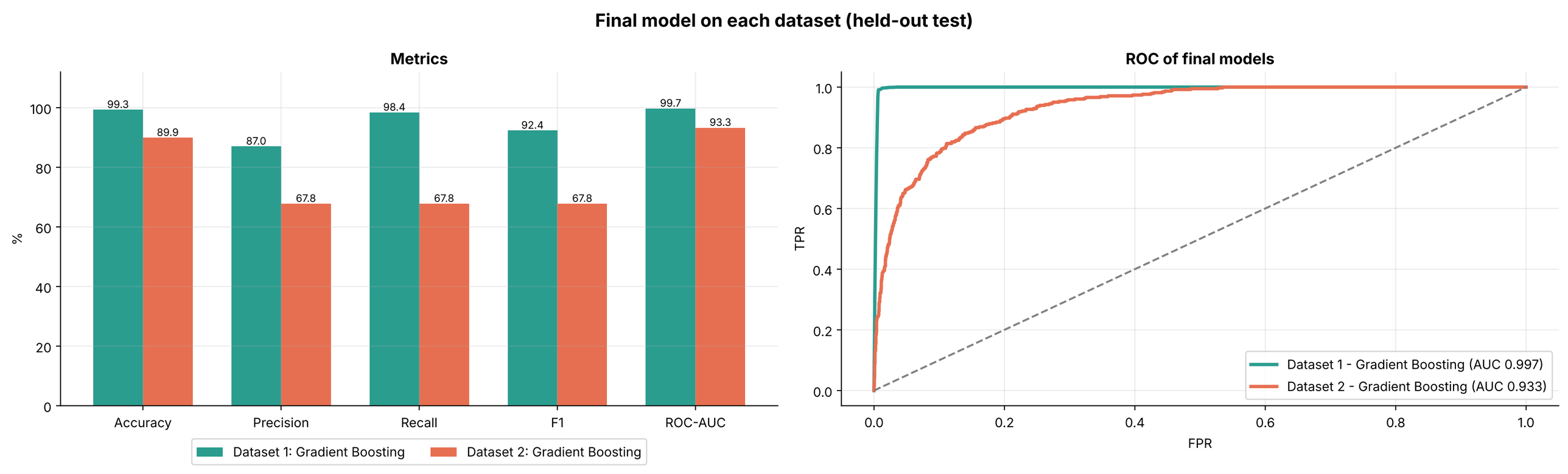
Dataset 2 (UCI Online Shoppers, 'Revenue') - final model: Gradient Boosting

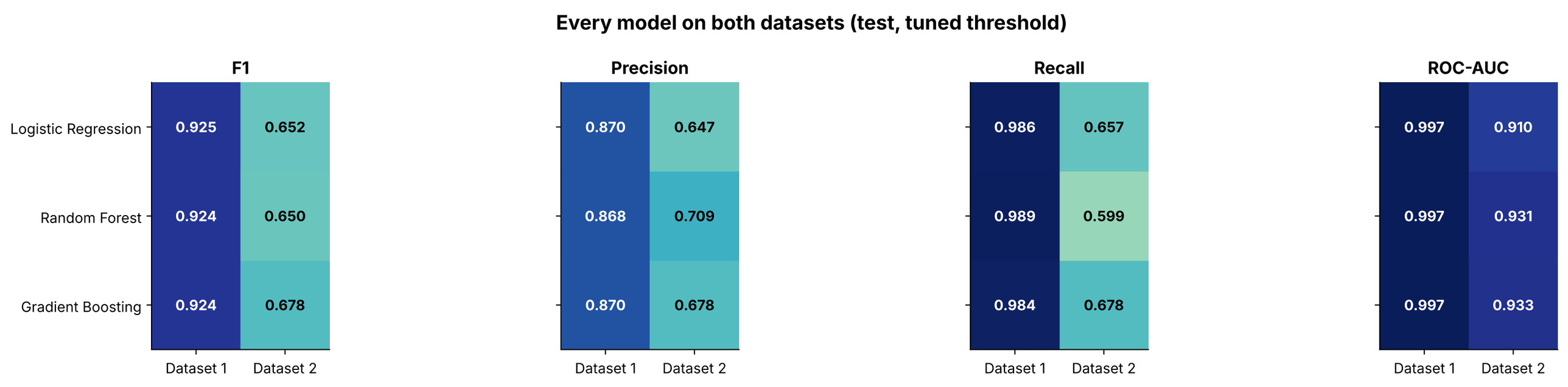


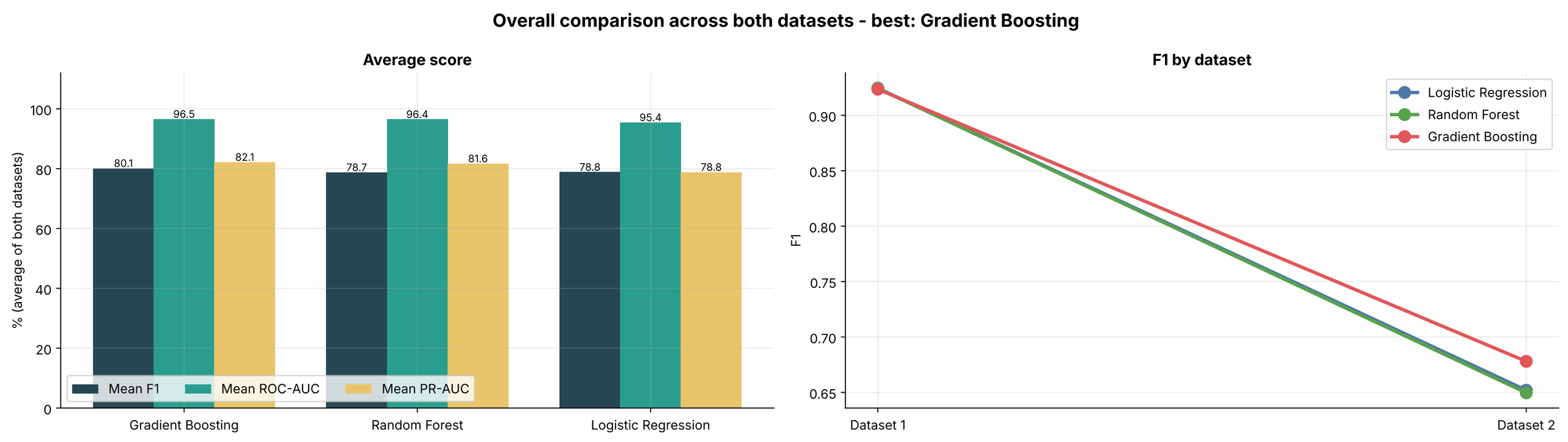


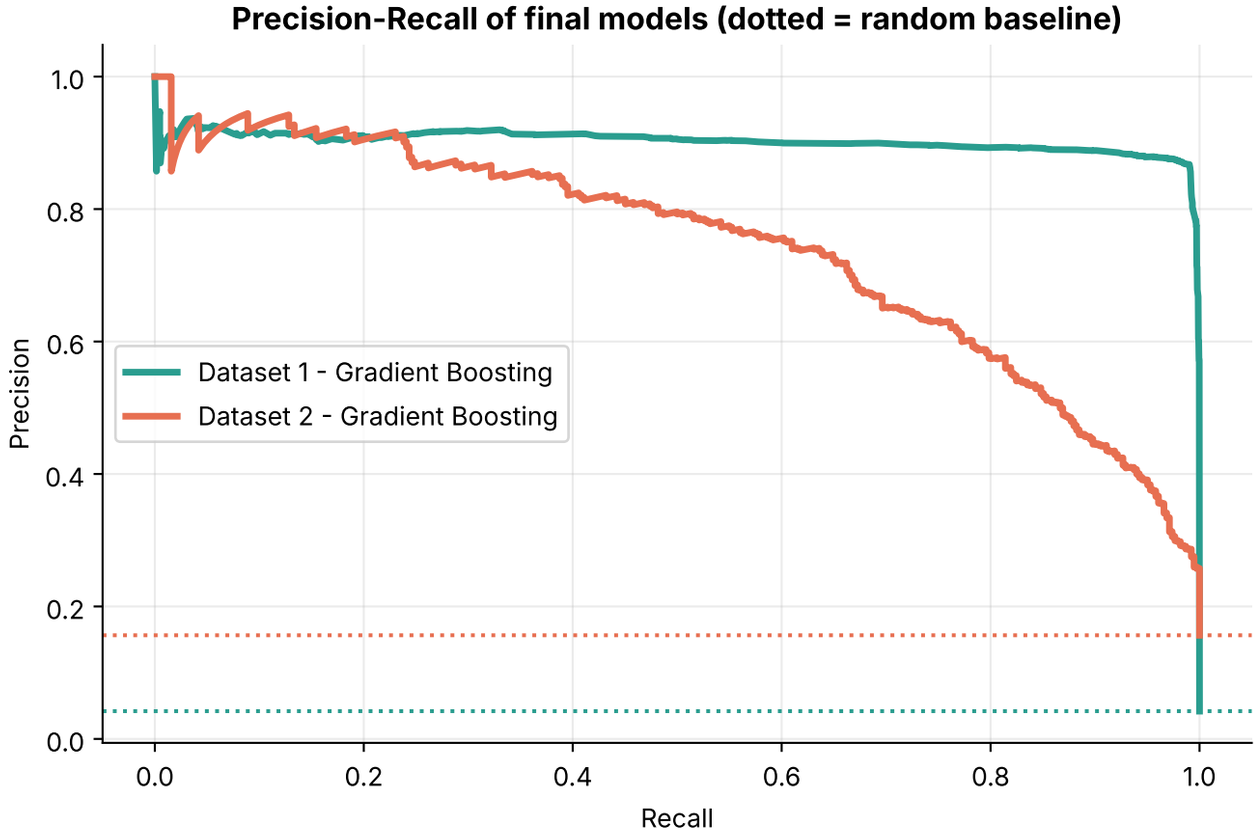


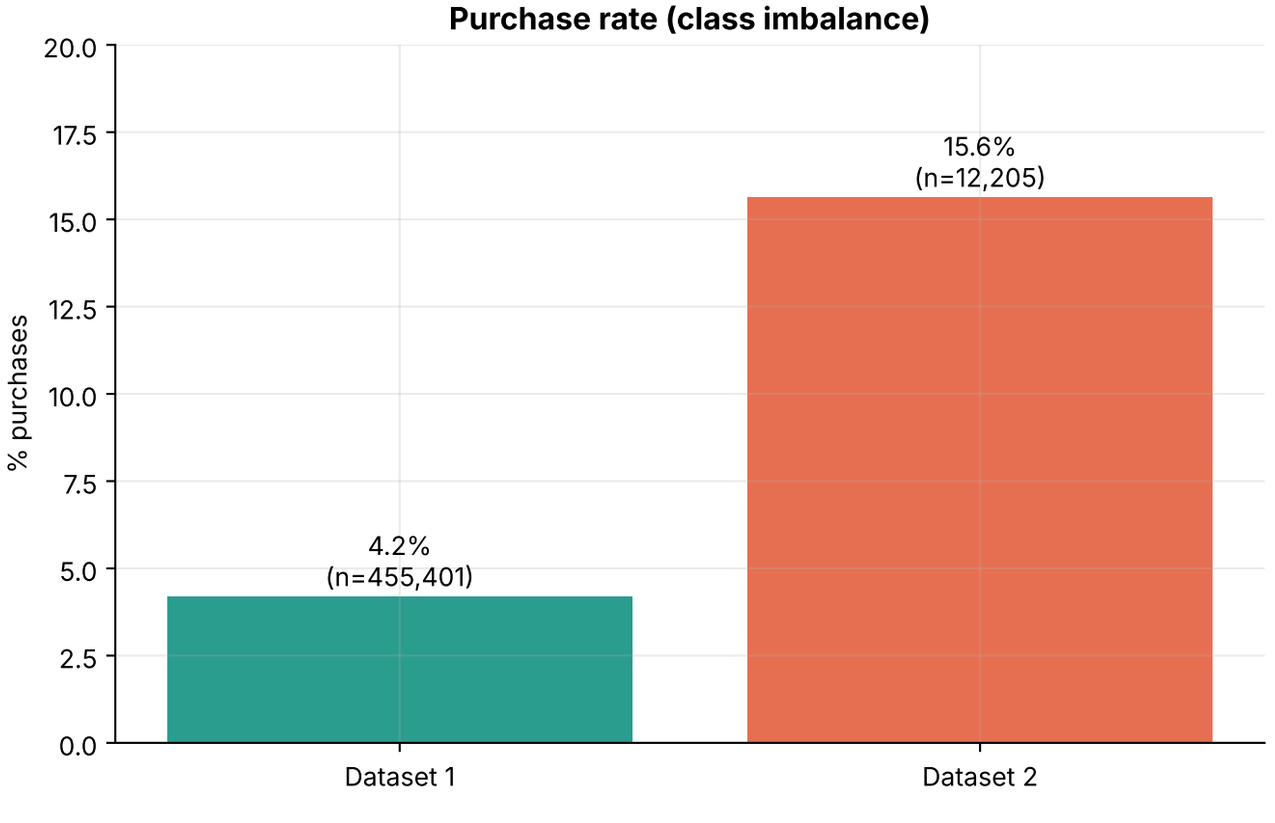

Dataset 2 (UCI Online Shoppers, 'Revenue')
Top-12 features (Gradient Boosting, permutation importance, drop in ROC-AUC)
PageValues
Month_May
ExitRates
Month_Mar
Month_Nov
AvgTimePerPage
ExitBounceDiff
PageValuePerProductPage
BounceRates
TotalDuration
TrafficType_3
ProductRelated_Duration
0.00
0.05
0.10
0.15
0.20
0.25
importance

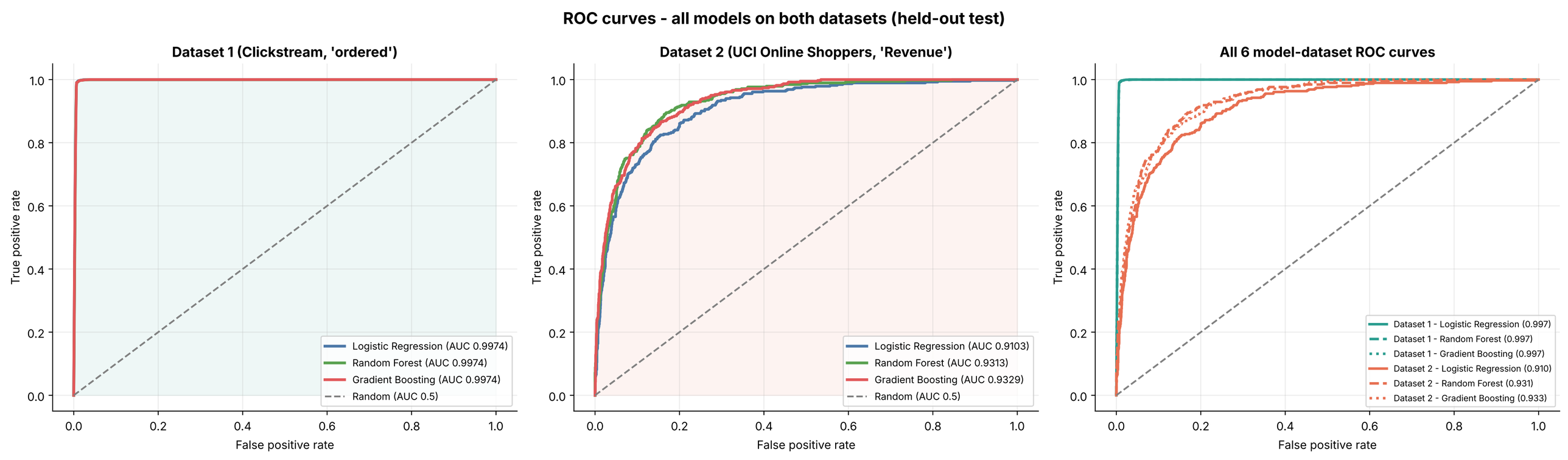

Final model on each dataset (held-out test)
Metrics
%
0
20
40
60
80
100
99.3
89.9
87.0
67.8
98.4
67.8
92.4
67.8
99.7
93.3
Accuracy
Precision
Recall
F1
ROC-AUC
Dataset 1: Gradient Boosting
Dataset 2: Gradient Boosting
ROC of final models
TPR
FPR
0.0
0.2
0.4
0.6
0.8
1.0
Dataset 1 - Gradient Boosting (AUC 0.997)
Dataset 2 - Gradient Boosting (AUC 0.933)

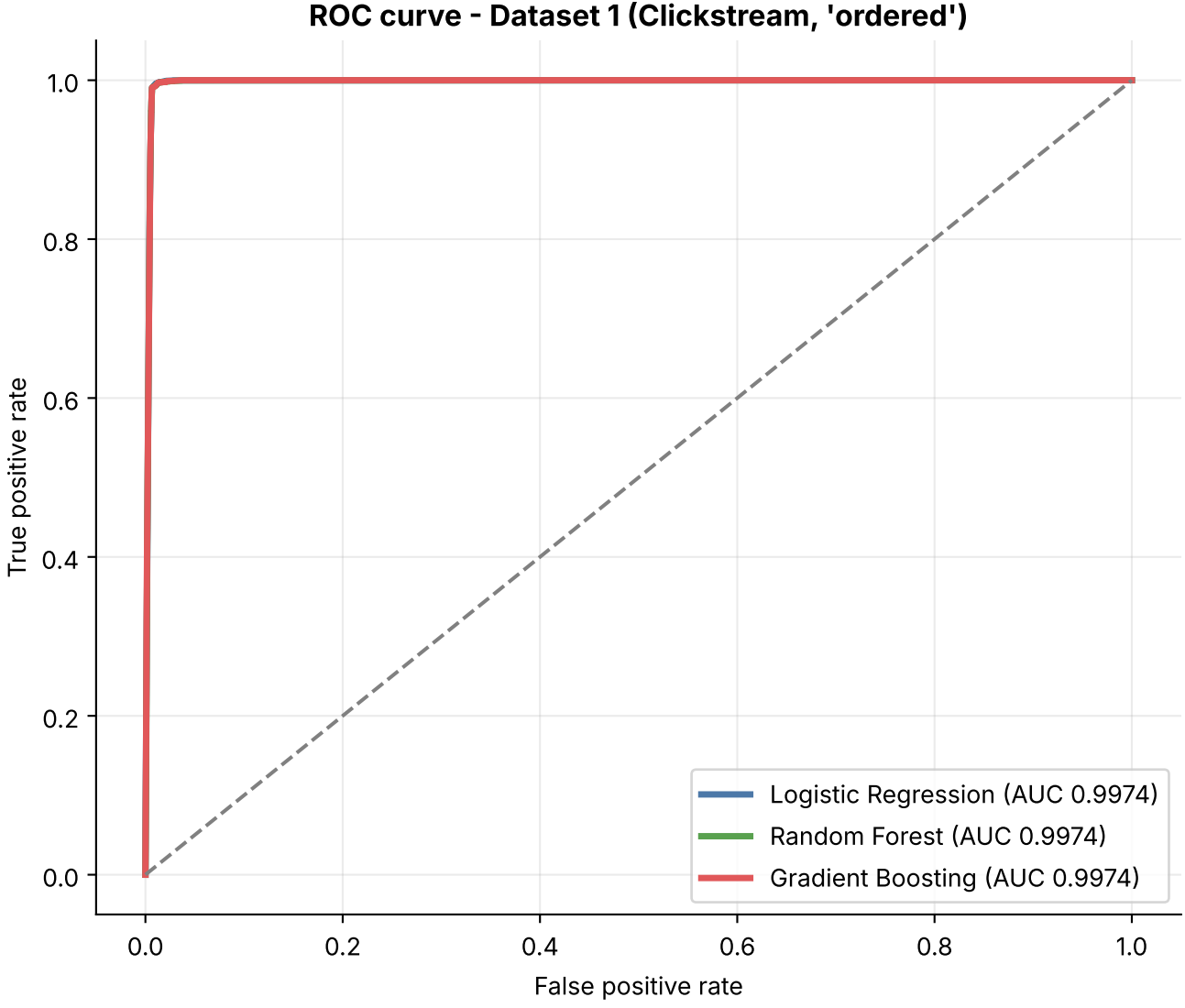

Every model on both datasets (test, tuned threshold)
F1
Precision
Recall
ROC-AUC
Logistic Regression
Random Forest
Gradient Boosting
0.925
0.652
0.924
0.650
0.924
0.678
0.870
0.647
0.868
0.709
0.870
0.678
0.986
0.657
0.989
0.599
0.984
0.678
0.997
0.910
0.997
0.931
0.997
0.933
Dataset 1
Dataset 2

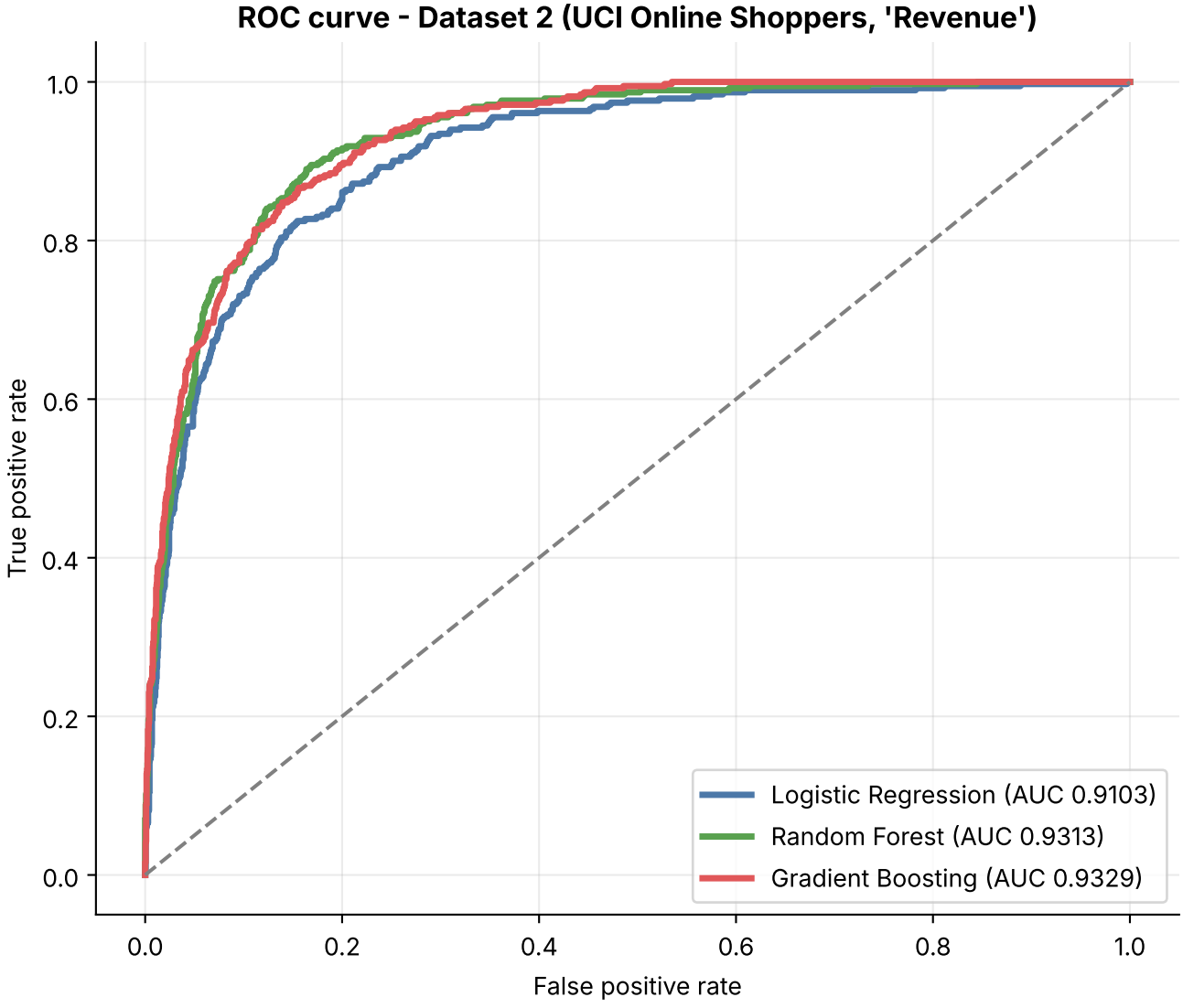

Overall comparison across both datasets - best: Gradient Boosting
Average score
% (average of both datasets)
0
20
40
60
80
100
80.1
96.5
82.1
78.7
96.4
81.6
78.8
95.4
78.8
Mean F1
Mean ROC-AUC
Mean PR-AUC
Gradient Boosting
Random Forest
Logistic Regression
F1 by dataset
F1
0.65
0.70
0.75
0.80
0.85
0.90
Logistic Regression
Random Forest
Gradient Boosting
Dataset 1
Dataset 2


Precision-Recall of final models (dotted = random baseline)
Precision
Recall
Dataset 1 - Gradient Boosting
Dataset 2 - Gradient Boosting
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0


Purchase rate (class imbalance)
% purchases
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
4.2%
(n=455,401)
15.6%
(n=12,205)
Dataset 1
Dataset 2

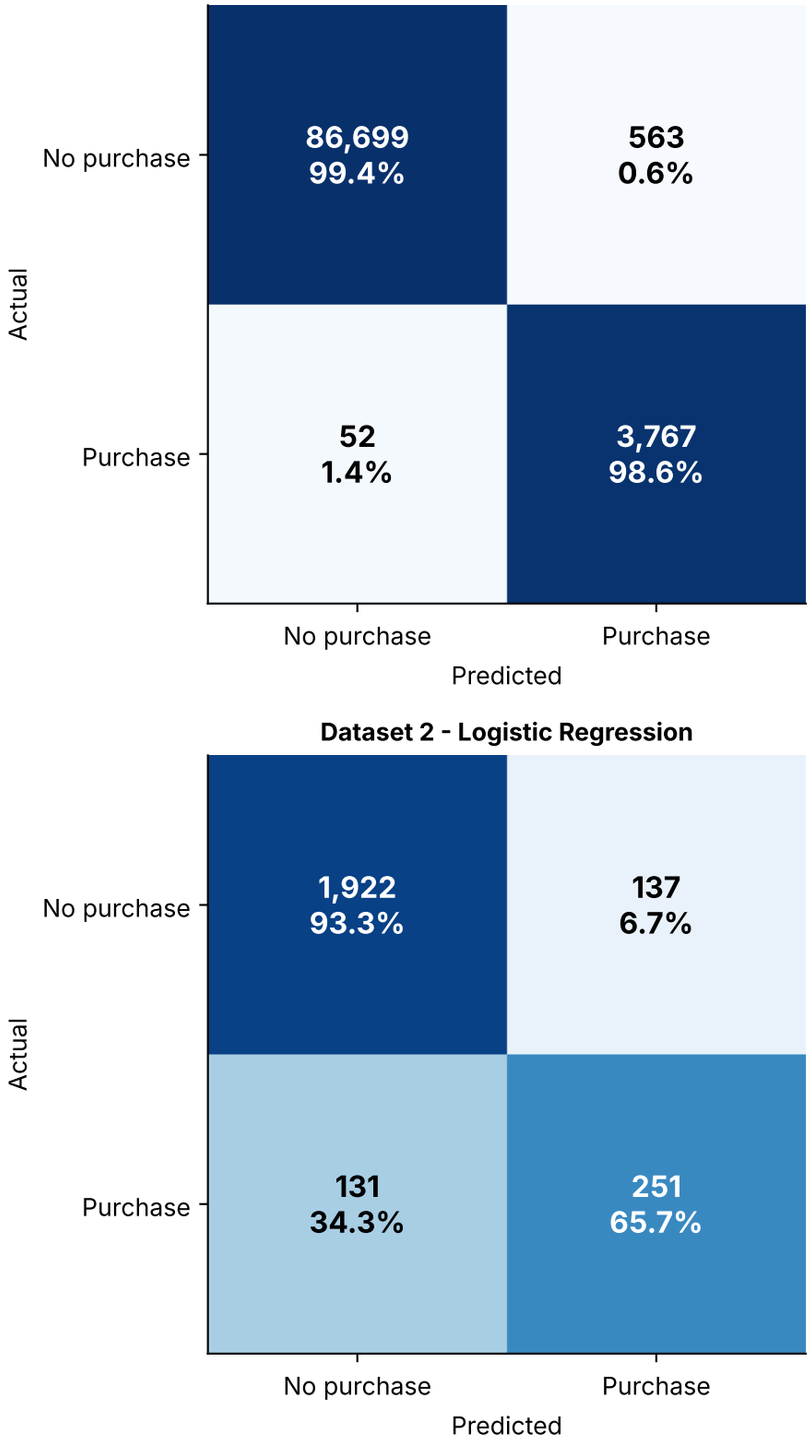

ROC curves - all models on both datasets (held-out test)
Dataset 1 (Clickstream, 'ordered')
Logistic Regression (AUC 0.9974)
Random Forest (AUC 0.9974)
Gradient Boosting (AUC 0.9974)
Random (AUC 0.5)
Dataset 2 (UCI Online Shoppers, 'Revenue')
Logistic Regression (AUC 0.9103)
Random Forest (AUC 0.9313)
Gradient Boosting (AUC 0.9329)
Random (AUC 0.5)
All 6 model-dataset ROC curves
Dataset 1 - Logistic Regression (0.997)
Dataset 1 - Random Forest (0.997)
Dataset 1 - Gradient Boosting (0.997)
Dataset 2 - Logistic Regression (0.910)
Dataset 2 - Random Forest (0.931)
Dataset 2 - Gradient Boosting (0.933)
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0


ROC curve - Dataset 1 (Clickstream, 'ordered')
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC 0.9974)
Random Forest (AUC 0.9974)
Gradient Boosting (AUC 0.9974)

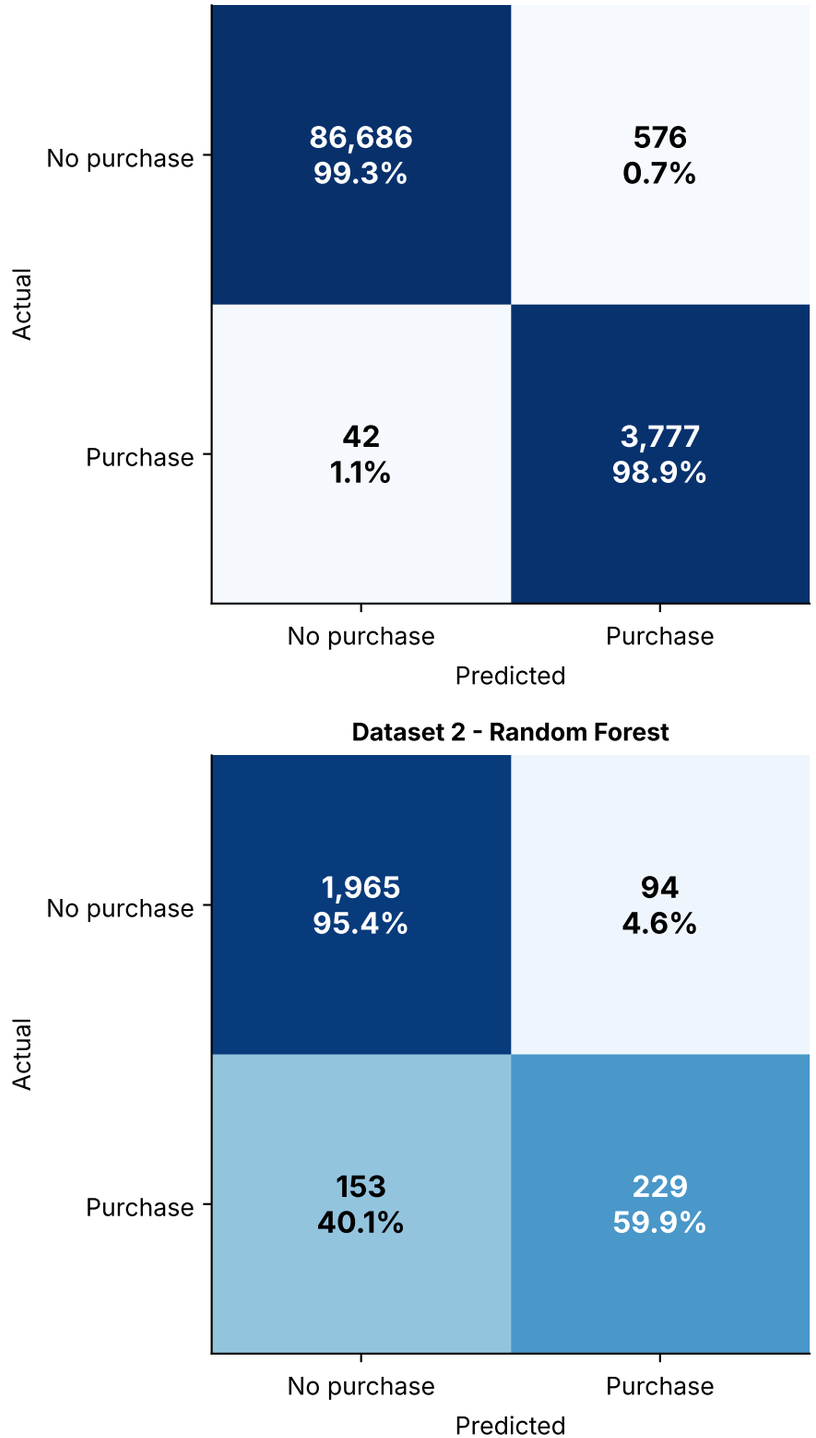

ROC curve - Dataset 2 (UCI Online Shoppers, 'Revenue')
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC 0.9103)
Random Forest (AUC 0.9313)
Gradient Boosting (AUC 0.9329)

Confusion matrices - every model on both datasets (test, tuned threshold; count and row %)

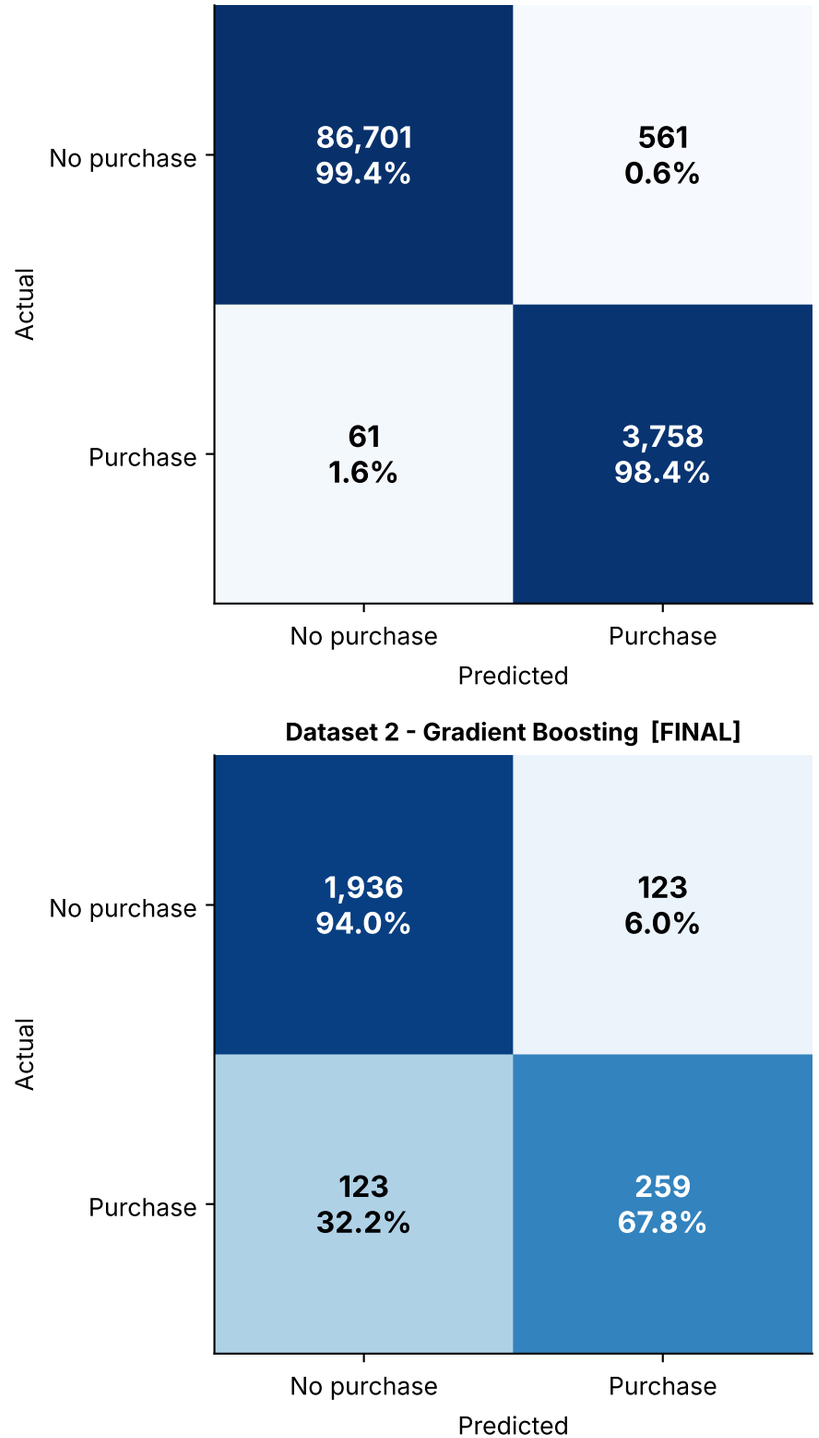
Dataset 1 - Logistic Regression



Dataset 1 - Random Forest



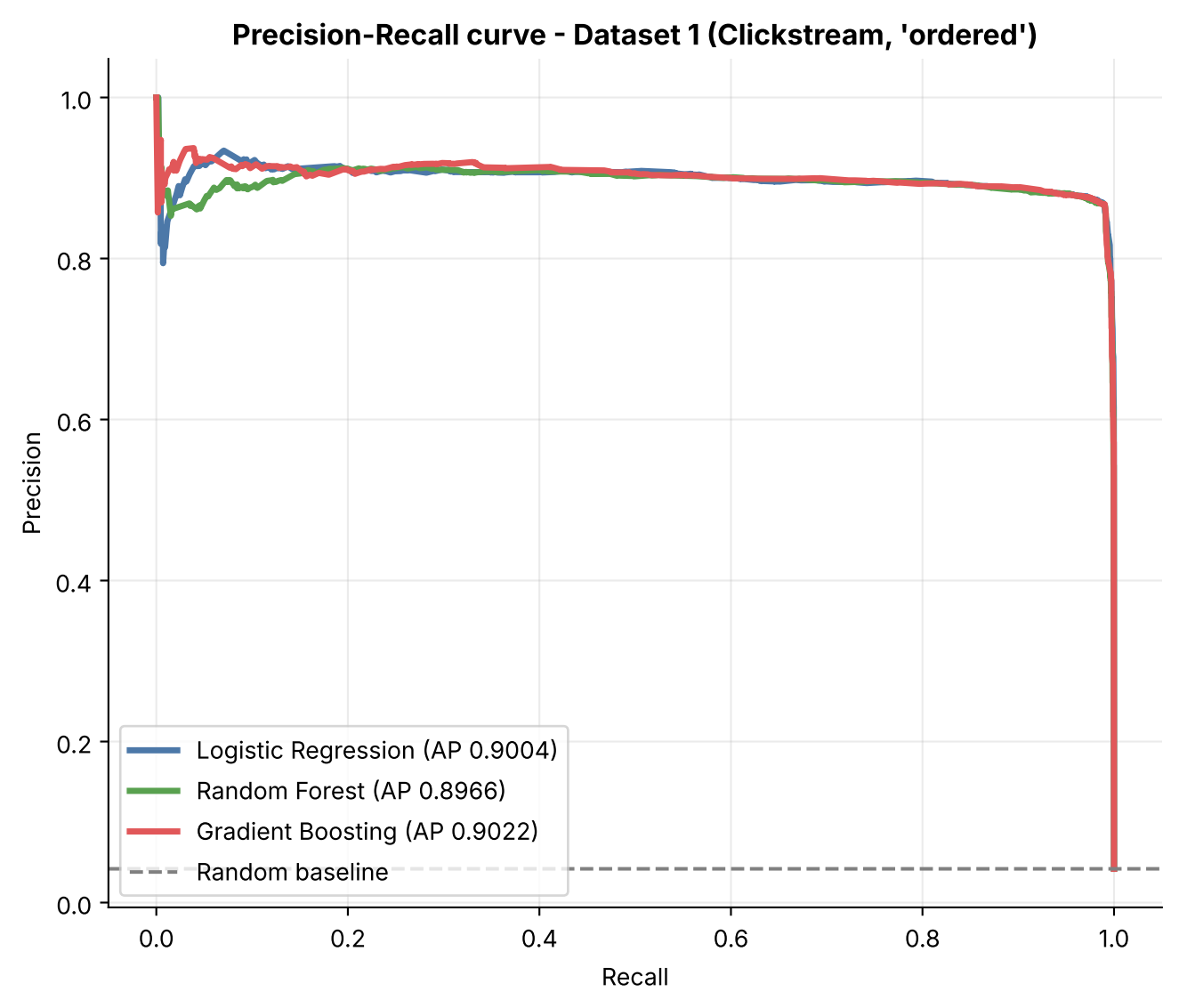
Dataset 1 - Gradient Boosting [FINAL]

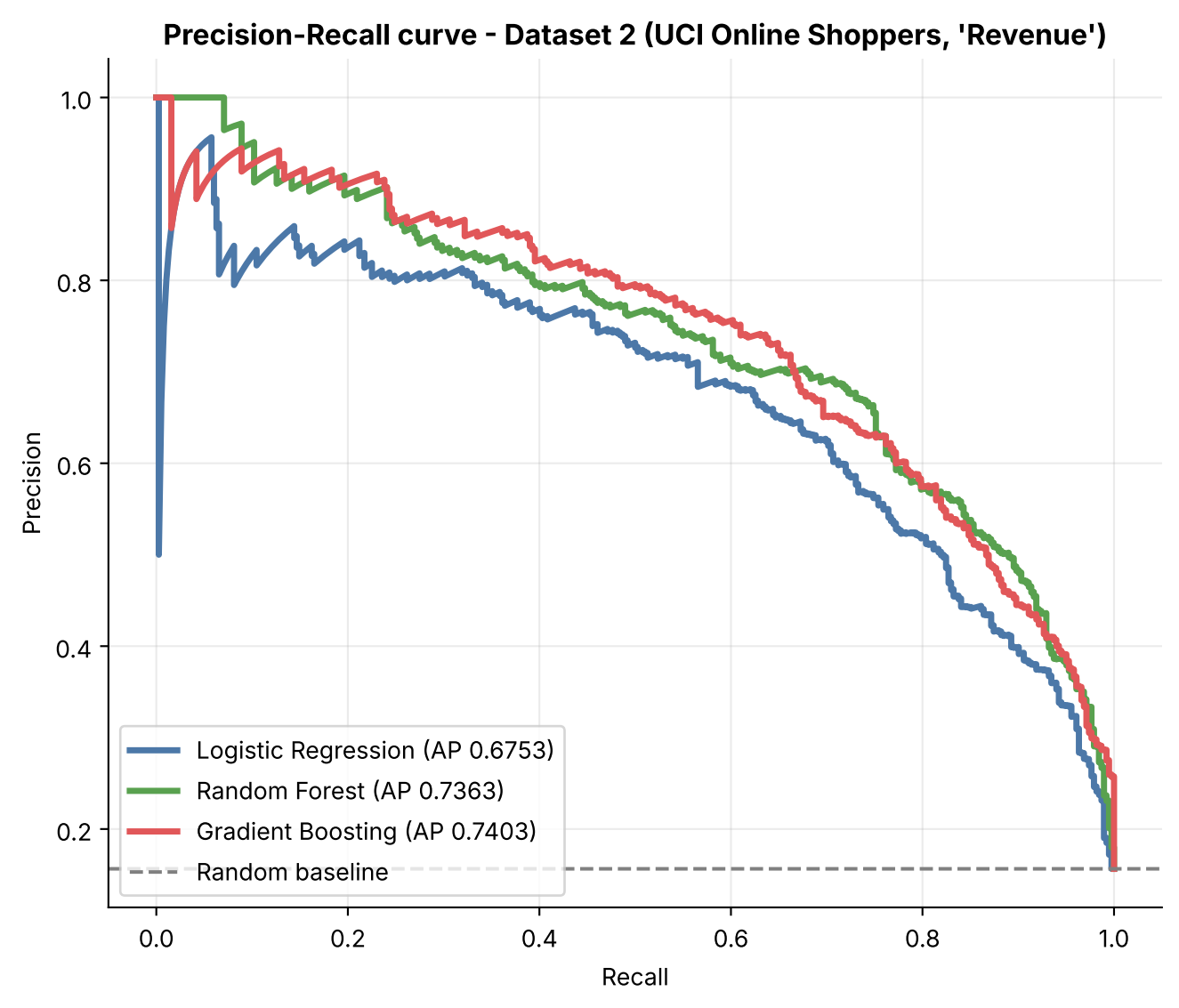


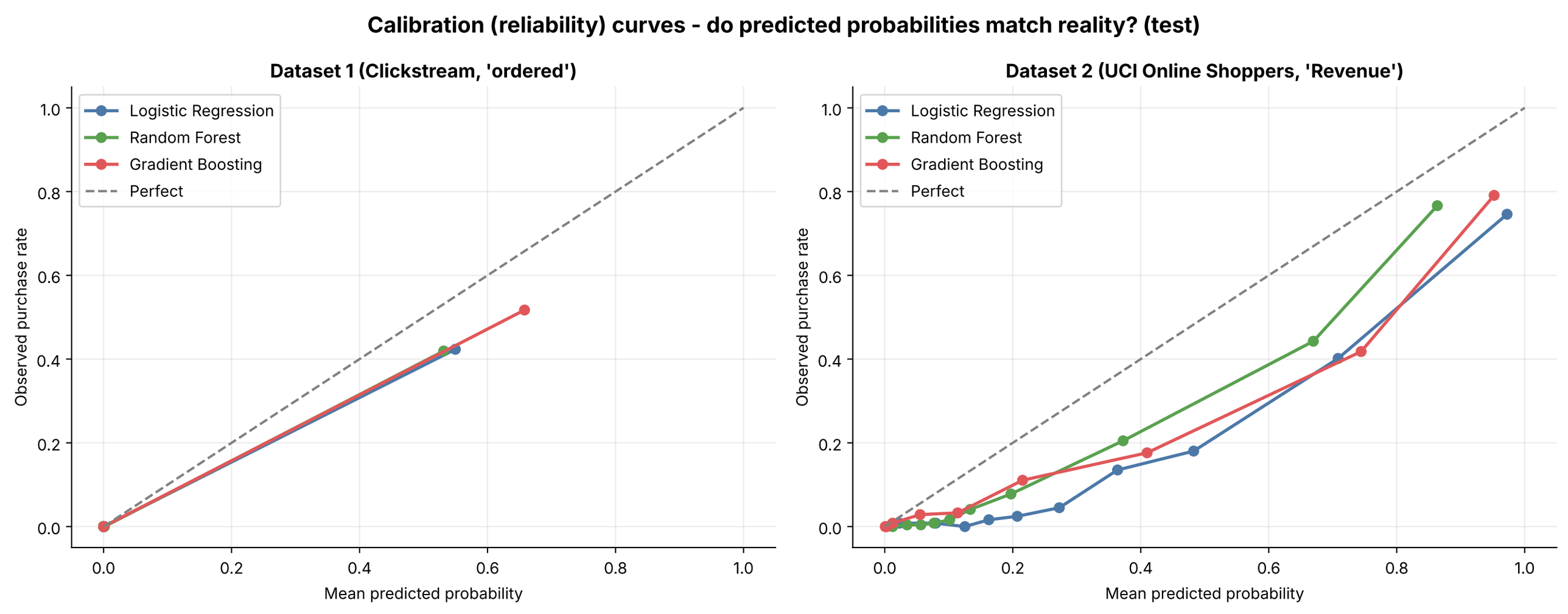
Dataset 2 - Logistic Regression

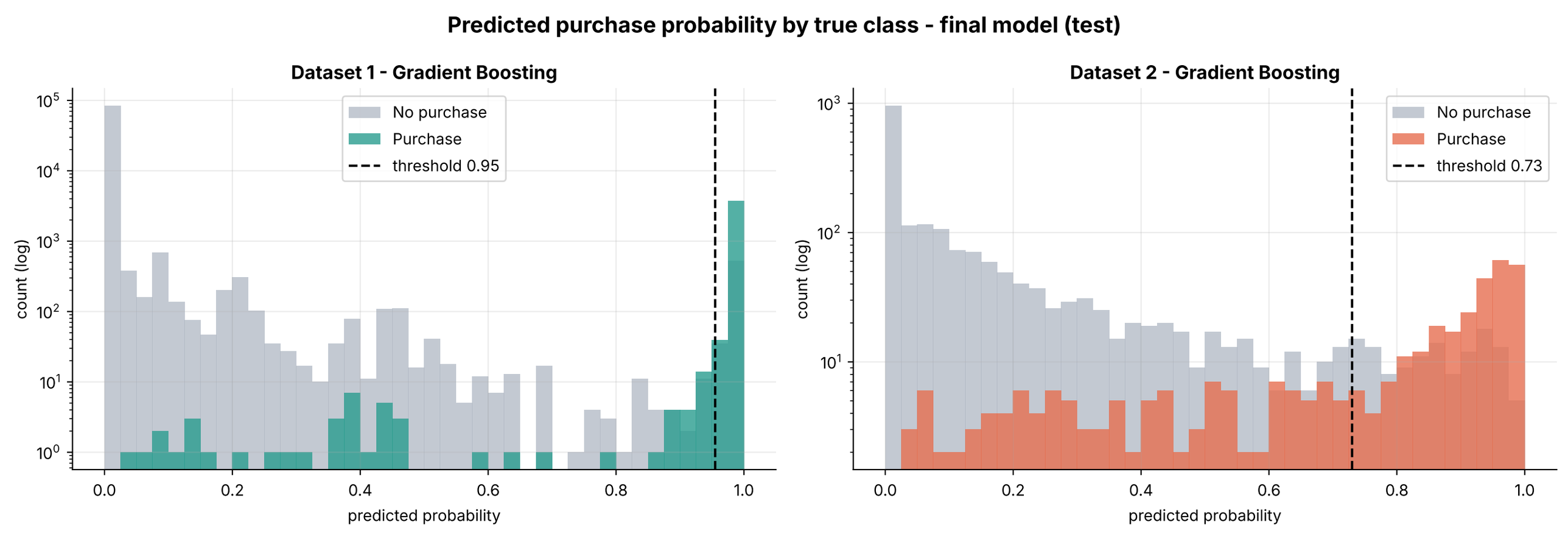
Dataset 2 - Random Forest

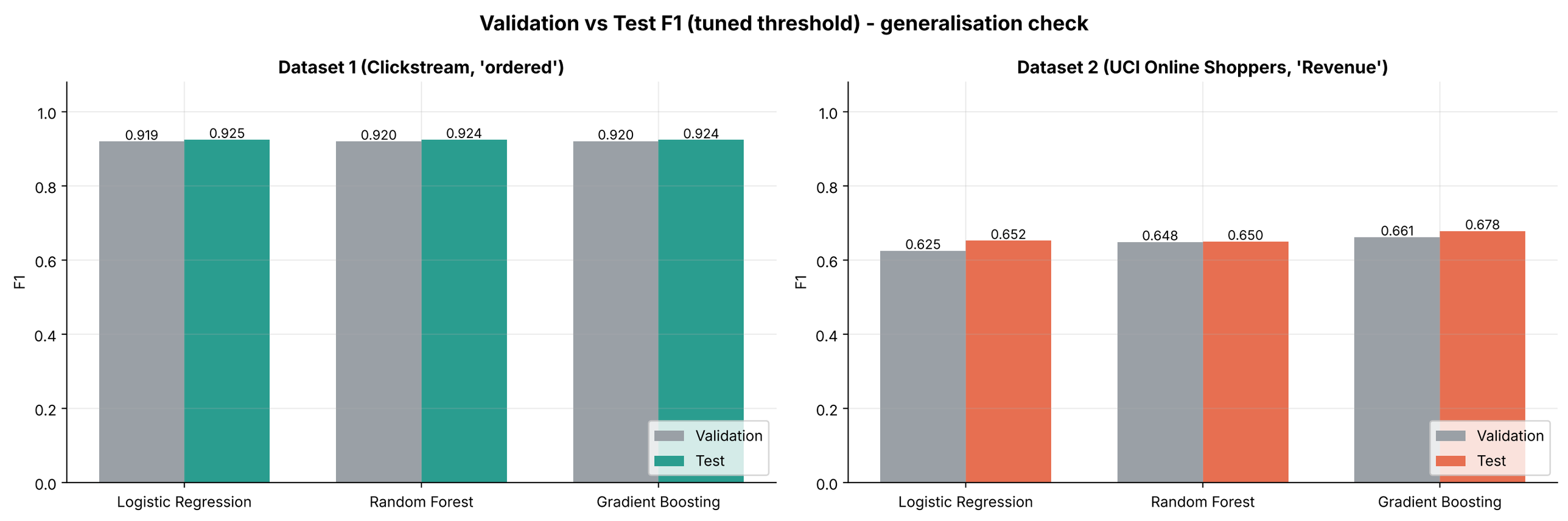
Dataset 2 - Gradient Boosting [FINAL]


Precision-Recall curve - Dataset 1 (Clickstream, 'ordered')
Precision
Recall
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AP 0.9004)
Random Forest (AP 0.8966)
Gradient Boosting (AP 0.9022)
Random baseline


Precision-Recall curve - Dataset 2 (UCI Online Shoppers, 'Revenue')
Precision
Recall
1.0
0.8
0.6
0.4
0.2
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AP 0.6753)
Random Forest (AP 0.7363)
Gradient Boosting (AP 0.7403)
Random baseline


Calibration (reliability) curves - do predicted probabilities match reality? (test)
Dataset 1 (Clickstream, 'ordered')
Dataset 2 (UCI Online Shoppers, 'Revenue')
Logistic Regression
Random Forest
Gradient Boosting
Perfect
Observed purchase rate
Mean predicted probability
0.0
0.2
0.4
0.6
0.8
1.0


Predicted purchase probability by true class - final model (test)
Dataset 1 - Gradient Boosting
No purchase
Purchase
threshold 0.95
count (log)
predicted probability
Dataset 2 - Gradient Boosting
No purchase
Purchase
threshold 0.73
count (log)
predicted probability


Validation vs Test F1 (tuned threshold) - generalisation check
Dataset 1 (Clickstream, 'ordered')
Dataset 2 (UCI Online Shoppers, 'Revenue')
F1
1.0
0.8
0.6
0.4
0.2
0.0
0.919
0.925
0.920
0.924
0.920
0.924
0.625
0.652
0.648
0.650
0.661
0.678
Logistic Regression
Random Forest
Gradient Boosting
Validation
Test